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Basic Trigonometry

Any triangle, three rules. Look at what you are given — it tells you which rule to use.

THE TOPIC ON ONE MAP

Basic Trigonometry

01 — Sine rule and the ambiguous case

an opposite pair

obtuse: $180^\circ - \theta$

two triangles

02 — Cosine rule

two sides + angle

three sides

03 — Area $\frac{1}{2}ab \sin C$

area given

work backwards

04 — Exact values, multi-step shapes

no calculator

shared side

Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

a side–angle **pair** is known**Cosine rule**

$$a^2 = b^2 + c^2 - 2bc \cos A$$

two sides & the angle between • or three sides

Area

$$\frac{1}{2}ab \sin C$$

two sides & the angle between

01 Sine Rule and the Ambiguous Case

Use it when you know one side **and its opposite angle**. For an angle, $\sin$ gives two answers: α and $180^\circ - \alpha$.

WHEN YOU SEE → YOU DO

“use the sine rule”, a side and its opposite angle

$$\rightarrow \frac{a}{\sin A} = \frac{b}{\sin B}$$

“angle is obtuse”, “the longest side”, “ $\theta > 90^\circ$ ”

$$\rightarrow \text{take } 180^\circ - \alpha$$

“two triangles”, “the other possible triangle”

$$\rightarrow \text{both } \alpha \text{ and } 180^\circ - \alpha$$

“to 4 decimal places” then “hence”

$$\rightarrow \text{keep the exact sine, then invert}$$

WORKED MODEL • JUNE 2022

$AB = 21$, $BC = 13$, $\angle BAC = 25^\circ$, $\angle ACB = x^\circ$; AB is the longest side.

$$1 \quad \frac{\sin x}{21} = \frac{\sin 25^\circ}{13} \Rightarrow \sin x = 0.6827$$

sine rule

$$2 \quad x = 43.05^\circ \text{ or } 136.95^\circ$$

two answers

ANSWER $x = 136.95$ (longest side $\Rightarrow$ largest angle)

IDEA 01

Practise: Sine Rule and the Ambiguous Case

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Side from the sine rule

P1 • TOPIC 9 2 marks

In triangle ABC , $BC = 10$ cm, $\angle A = 70^\circ$, $\angle B = 40^\circ$. Find AC .

$AC =$ _____

Q2 **DR ESLAM** Two possible angles

P1 • TOPIC 9 3 marks

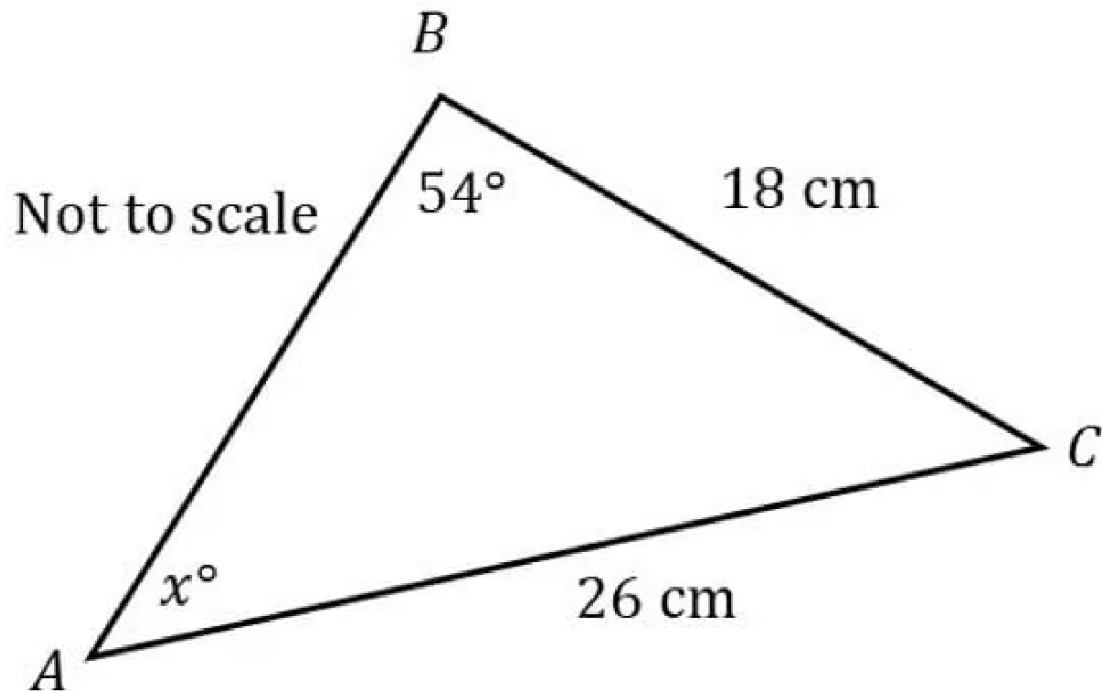
$AB = 8$, $BC = 6$, $\angle A = 40^\circ$. Find the two possible values of $\angle C$.

SELECTED PRACTICE

Q3 PRACTICE Practice · Easy

P1 · TOPIC 9 3 marks

6



x is an acute angle. Use the sine rule to calculate the angle x° . Give your answer to the nearest degree.

(3 marks)

Q4**PRACTICE**

Practice • Medium

P1 • TOPIC 9 4 marks

5 Two triangles both have sides $AB = 5.3$ cm, $BC = 6.4$ cm and angle $ACB = 38^\circ$.

(i) Show that the angle BAC for one of the triangles is 132° , to 3 s.f.

(ii) Find the angle ABC for the other triangle.

(4 marks)**Q5****PRACTICE**

Practice • Hard

P1 • TOPIC 9 4 marks

5 Two triangles both have sides $AB = 2.8$ cm, $BC = 3.5$ cm and angle $ACB = 43^\circ$.

Find the largest angle in each of the triangles.

(4 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA11 · June 2022 · Q2

P1 · TOPIC 9 5 marks

2. In the triangle ABC ,

- $AB = 21$ cm
- $BC = 13$ cm
- angle $BAC = 25^\circ$
- angle $ACB = x^\circ$

(a) Use the sine rule to find the value of $\sin x^\circ$, giving your answer to 4 decimal places.
(2)

Given also that AB is the longest side of the triangle,

(b) find the value of x , giving your answer to 2 decimal places.
(3)

(Total 5 marks)

02 Cosine Rule

Two sides and the angle between them → the third side. Three sides → any angle (the largest is opposite the longest side).

WHEN YOU SEE → YOU DO

“find the length of BD ”, two sides and the angle between

$$\longrightarrow a^2 = b^2 + c^2 - 2bc \cos A$$

three sides, “find the largest angle”

→ angle opposite the longest side

“sides in the ratio 2:4:5”

→ use 2, 4, 5 for the angle

“show that $p^2 + q^2 - pq = 8$ ”

$$\longrightarrow \text{cosine rule with } \cos 60^\circ = \frac{1}{2}$$

bearings

→ draw north lines, find the angle inside the triangle

WORKED MODEL • OCTOBER 2025

Sides in the ratio 2 : 4 : 5; area 140 cm^2 .

$$1 \quad \cos \theta = \frac{4 + 16 - 25}{2(2)(4)} = -\frac{5}{16}$$

largest angle

$$2 \quad \theta = 108.2^\circ$$

invert

$$3 \quad \frac{1}{2}(2k)(4k) \sin 108.2^\circ = 140 \Rightarrow k = 6.07$$

area with $2k, 4k$

ANSWER 108.2° , shortest side 12.1 cm

IDEA 02

Practise: Cosine Rule

My questions first, then selected practice, then real past papers.

Q7 DR ESLAM Cosine rule, exact

P1 · TOPIC 9 2 marks

$AB = 8$, $AC = 5$, $\angle A = 60^\circ$. Find BC exactly.

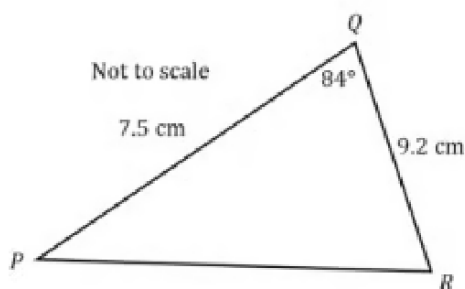
$BC =$ _____

SELECTED PRACTICE**Q8** PRACTICE Practice · Easy

P1 · TOPIC 9 3 marks

- 5 PQR is a triangle with angle $PQR = 84^\circ$ and side lengths $PQ = 7.5$ cm and $QR = 9.2$ cm.

Use cosine rule to calculate the length of PR .



(3 marks)

Q9**PRACTICE**

Practice · Medium

P1 · TOPIC 9 4 marks

- 1 (a) A triangle ABC has side lengths $AB = 5$ cm, $BC = 4$ cm and $AC = 7$ cm.

Calculate the size of the angle BAC , giving your answer to the nearest degree.

(2 marks)

- (b) A triangle PQR has side lengths $PQ = 12$ cm, $QR = 8$ cm and angle $PQR = 60^\circ$.

Calculate the length of RP , giving your answer to three significant figures.

(2 marks)**Q10****PRACTICE**

Practice · Medium

P1 · TOPIC 9 3 marks

- 7 Three telephone masts are used to triangulate locations. Mast B is due west of Mast A . Mast C is 0.6 km from Mast A and 0.7 km from Mast B on a bearing of 136° .

Show that the bearing of Mast C from Mast A is 213° .

(3 marks)

PAST PAPERS

Q11 PAST PAPER Pearson IAL WMA11 · October 2019 · Q4

P1 · TOPIC 9 5 marks

4. A parallelogram $ABCD$ has area 40 cm^2

Given that AB has length 10 cm , BC has length 6 cm and angle DAB is obtuse, find

- (a) the size of angle DAB , in degrees, to 2 decimal places, (3)

- (b) the length of diagonal BD , in cm , to one decimal place. (2)

(Total 5 marks)

Q12**PAST PAPER**

Pearson IAL WMA11 · October 2025 · Q3

P1 · TOPIC 9 5 marks

3. The sides of a triangle are in the ratio 2:4:5

(a) Find the size of the **largest** angle of the triangle, in degrees, to one decimal place.

(2)

Given that the area of the triangle is 140cm^2

(b) find the length of the **shortest** side, in cm, to one decimal place.

(3)

(Total for Question 3 is 5 marks)

03 Area $\frac{1}{2}ab \sin C$ — and Working Backwards

Area with the angle **between** the two sides. Given the area, the same formula finds $\sin C$ — which has two angles.

WHEN YOU SEE → YOU DO

“the area is 100 cm^2 , find $\sin \theta$ ”

$$\rightarrow \frac{1}{2}ab \sin \theta = 100$$

“given $\theta > 90^\circ$ ” after $\sin \theta$

$\rightarrow \cos \theta$ is negative in the cosine rule

parallelogram

$\rightarrow \text{area} = ab \sin \theta$ (two triangles)

“show that one possible value is 48.1° ”

$\rightarrow$ the other is $180^\circ - 48.1^\circ$

WORKED MODEL • JANUARY 2024

$AB = 15$, $AC = 25$, area 100 , $\theta > 90^\circ$.

$$1 \quad \frac{1}{2}(15)(25) \sin \theta = 100 \Rightarrow \sin \theta = \frac{8}{15}$$

area formula

$$2 \quad \cos \theta = -\sqrt{1 - \frac{64}{225}}$$

obtuse: negative

$$3 \quad BC^2 = 225 + 625 + 2(15)(25)(0.8433)$$

cosine rule

ANSWER $BC = 38.5 \text{ cm}$

IDEA 03

Practise: Area and Working Backwards

My questions first, then selected practice, then real past papers.

Q13 DR ESAM Area

P1 • TOPIC 9 2 marks

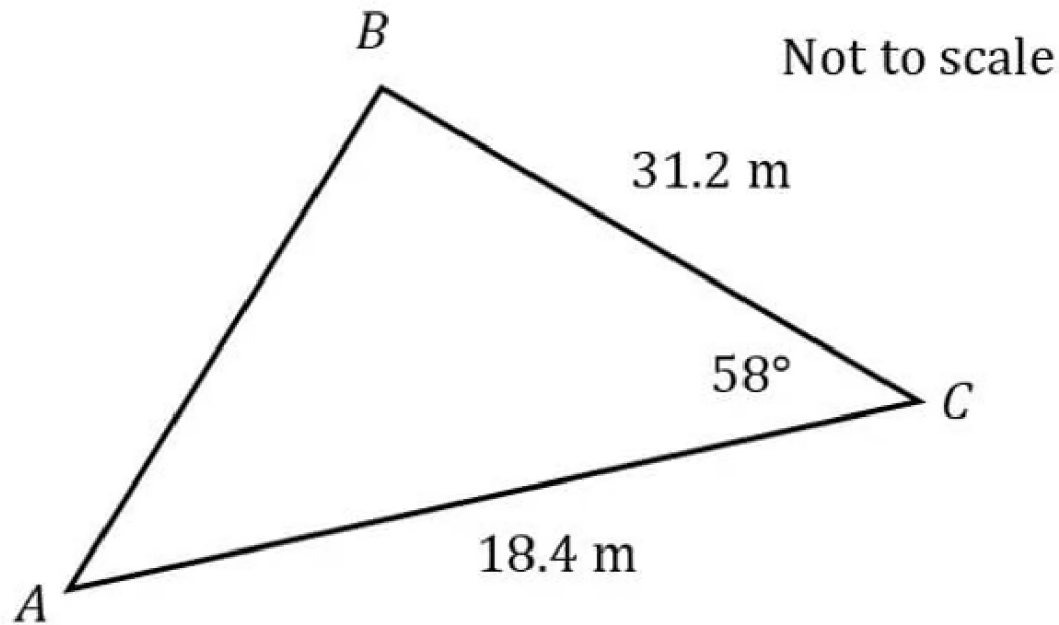
Find the area of a triangle with sides 6 cm and 9 cm and included angle 150° .

SELECTED PRACTICE

Q14 PRACTICE Practice · Easy

P1 · TOPIC 9 3 marks

7 A triangular field is shown in the diagram below. Calculate the area of the field, give your answer to the nearest square metre.



(3 marks)

Q15**PRACTICE**

Practice · Easy

P1 · TOPIC 9 3 marks

- 9 The area of a triangle WXY is 75 cm^2 . $XY = 14.3 \text{ cm}$ and angle $WXY = 104^\circ$.

Using the formula $A = \frac{1}{2} ab \sin C$, calculate the length of WX . Give your answer to three significant figures.

(3 marks)

Q16**PRACTICE**

Practice · Medium

P1 · TOPIC 9 8 marks

- 8 (a)** A triangular field has an area of 10m^2 . The field has two side lengths of 3.2 m and 8.4 m which have an angle θ between them.

Show that one possible value for θ is 48.1° to three significant figures.

(3 marks)

- (b)** Show that there is another possible value that works for θ . You must show all your working.

(2 marks)

- (c)** Using $\theta = 48.1^\circ$. Calculate the perimeter of the field. Giving your answer to three significant figures.

(3 marks)

PAST PAPERS

Q17 PAST PAPER Pearson IAL WMA11 • January 2024 • Q2

P1 • TOPIC 9 5 marks

2. The triangle ABC is such that

- $AB = 15$ cm
- $AC = 25$ cm
- angle $BAC = \theta^\circ$
- area triangle $ABC = 100$ cm²

(a) Find the value of $\sin \theta^\circ$

(2)

Given that $\theta > 90$ (b) find the length of BC , in cm, to 3 significant figures.

(3)

(Total for Question 2 is 5 marks)

Q18 **PAST PAPER** Pearson IAL WMA11 · June 2026 · Q3**P1 · TOPIC 9** **6 marks**

3. A triangular plot of land, ABC , has area 60 m^2

Given

- $AB = 15 \text{ m}$
- $AC = 10 \text{ m}$
- angle $BAC = \theta^\circ$

(a) show that $\sin \theta^\circ = \frac{4}{5}$ (1)

Given also that θ° is the **largest** angle in the triangle,

(b) find the length of the side BC , in metres, to 3 significant figures, (3)

(c) find the size of the **smallest** angle in the triangle, in degrees, to one decimal place. (2)

Question 3 continued

(Total for Question 3 is 6 marks)

04 Exact Values and Multi-Step Shapes

“Show all stages”, “exact” — use $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$ and keep surds. Frames and plots join two triangles along a shared side.

WHEN YOU SEE → YOU DO

“solutions relying on calculator ... not acceptable”

→ exact values, algebra

“AC is 2 metres longer than AB”

→ $q = p + 2$, substitute

two triangles sharing a side

→ solve the first, carry the shared side

	30°	45°	60°
sin	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
cos	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$

WORKED MODEL • JUNE 2021

$$BC = 2\sqrt{2}, \angle A = 60^\circ, AB = p, AC = q = p + 2.$$

$$1 \quad 8 = p^2 + q^2 - 2pq \cdot \frac{1}{2}$$

cosine rule, exact

$$2 \quad p^2 + 2p - 4 = 0 \Rightarrow p = -1 + \sqrt{5}$$

substitute $q = p + 2$

ANSWER $p = \sqrt{5} - 1, q = \sqrt{5} + 1, \text{ area} = \frac{1}{2}pq \cdot \frac{\sqrt{3}}{2} = \sqrt{3}$

IDEA 04

Practise: Exact Values and Multi-Step Shapes

My questions first, then real past papers.

Q19**DR ESLAM**

Exact and algebraic

P1 · TOPIC 9

4 marks

$AB = x$, $AC = x + 1$, $BC = \sqrt{7}$, $\angle A = 60^\circ$. Find x .

$x =$ _____

PAST PAPERS

Q20 PAST PAPER Pearson IAL WMA11 · June 2021 · Q3

P1 · TOPIC 9 9 marks

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

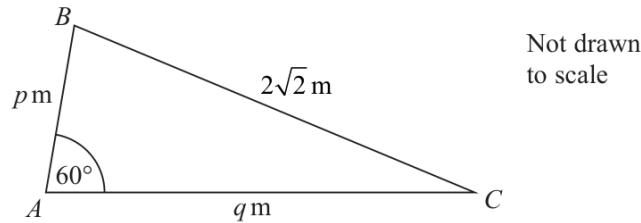
**Figure 1**

Figure 1 shows the plan view of a flower bed.
The flowerbed is in the shape of a triangle ABC with

- $AB = p$ metres
- $AC = q$ metres
- $BC = 2\sqrt{2}$ metres
- angle $BAC = 60^\circ$

(a) Show that

$$p^2 + q^2 - pq = 8 \quad (2)$$

Given that side AC is 2 metres longer than side AB , use algebra to find

- (b) (i) the exact value of p ,
- (ii) the exact value of q . (5)

Using the answers to part (b),

- (c) calculate the exact area of the flower bed. (2)

(Total 9 marks)

Q21 PAST PAPER Pearson IAL WMA11 • October 2023 • Q5

P1 • TOPIC 9 7 marks

5.

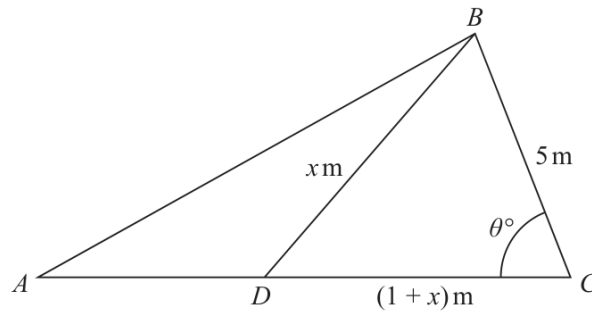
Diagram NO
accurately dr**Figure 2**

Figure 2 shows the plan view of a frame for a flat roof.

The shape of the frame consists of triangle ABD joined to triangle BCD .

Given that

- $BD = x$ m
- $CD = (1 + x)$ m
- $BC = 5$ m
- angle $BCD = \theta^\circ$

(a) show that $\cos \theta^\circ = \frac{13 + x}{5 + 5x}$

(2)

Given also that

- $x = 2\sqrt{3}$
- angle $BAC = 30^\circ$
- ADC is a straight line

(b) find the area of triangle ABC , giving your answer, in m^2 , to one decimal place.

(5)

(Total for Question 5 is 7 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------------|---|
| 01 | Sine rule | A side with its opposite angle. Angle: α or $180^\circ - \alpha$. |
| 02 | Cosine rule | Two sides + included angle, or three sides.
Largest angle opposite longest side. |
| 03 | Area | $\frac{1}{2}ab \sin C$. Given area $\rightarrow \sin C \rightarrow$ two possible angles. |
| 04 | Exact step / multi- | $\sin 60 = \frac{\sqrt{3}}{2}, \cos 60 = \frac{1}{2}$. Shared side carries across. |

BOTTOM LINE Pair known $\rightarrow$ sine rule • included angle or three sides $\rightarrow$ cosine rule • always check the obtuse answer.

Answers

Basic Trigonometry • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • SINE RULE AND THE AMBIGUOUS CASE

- Q1** Dr Eslam • Side from the sine rule

$$\frac{10 \sin 40^\circ}{\sin 70^\circ} = 6.84 \text{ cm}$$
- Q2** Dr Eslam • Two possible angles

$$\sin C = \frac{8 \sin 40^\circ}{6} = 0.857: C = 59.0^\circ \text{ or } 121.0^\circ$$
- Q3** Practice • Easy

$$x = 34^\circ$$
- Q4** Practice • Medium
 (i) $\sin A = \frac{6.4 \sin 38^\circ}{5.3}: 132^\circ$ (ii) $\angle ABC = 94.0^\circ$
- Q5** Practice • Hard

$$\sin A = \frac{3.5 \sin 43^\circ}{2.8} = 0.853: A = 58.5^\circ \text{ or } 121.5^\circ;$$

 largest angles 78.5° and 121.5°
- Q6** Pearson IAL WMA11 • June 2022 • Q2
 (a) 0.6827 (b) $x = 136.95$

IDEA 02 • COSINE RULE

- Q7** Dr Eslam • Cosine rule, exact

$$\sqrt{64 + 25 - 40} = 7$$
- Q8** Practice • Easy

$$PR = 11.2 \text{ cm}$$
- Q9** Practice • Medium
 (a) 34° (b) 10.6 cm
- Q10** Practice • Medium
 bearing 213° (shown)
- Q11** Pearson IAL WMA11 • October 2019 • Q4
 (a) 138.19° (b) 15.0 cm
- Q12** Pearson IAL WMA11 • October 2025 • Q3
 (a) 108.2° (b) 12.1 cm

IDEA 03 • AREA AND WORKING BACKWARDS

- Q13** Dr Eslam • Area 13.5 cm^2
- Q14** Practice • Easy 243 m^2
- Q15** Practice • Easy $WX = 10.8 \text{ cm}$
- Q16** Practice • Medium (b) 131.9° also works (c) 18.3 m
- Q17** Pearson IAL WMA11 • January 2024 • Q2 (a) $\frac{8}{15}$ (b) 38.5 cm
- Q18** Pearson IAL WMA11 • June 2026 • Q3 (b) 22.5 m (c) 20.9°

IDEA 04 • EXACT VALUES AND MULTI-STEP SHAPES

- Q19** Dr Eslam • Exact and algebraic $7 = x^2 + (x+1)^2 - x(x+1) \Rightarrow x^2 + x - 6 = 0: x = 2$
- Q20** Pearson IAL WMA11 • June 2021 • Q3 (b) $p = \sqrt{5} - 1, q = \sqrt{5} + 1$ (c) $\sqrt{3} \text{ m}^2$
- Q21** Pearson IAL WMA11 • October 2023 • Q5 (b) 16.1 m^2